Deployment of 3-Tier Application Using AWS Services

Submitted to - Vikul Sir

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Project Overview

This project demonstrates the manual deployment of a traditional 3-tier web application architecture on AWS.

The application consists of:

1. Frontend Layer (Presentation Tier)
2. Backend Layer (Application Tier)
3. Database Layer (Database Tier)

GitHub Repository:
https://github.com/rahulnalathati-21/AWS-3-Tier-Architecture/tree/main

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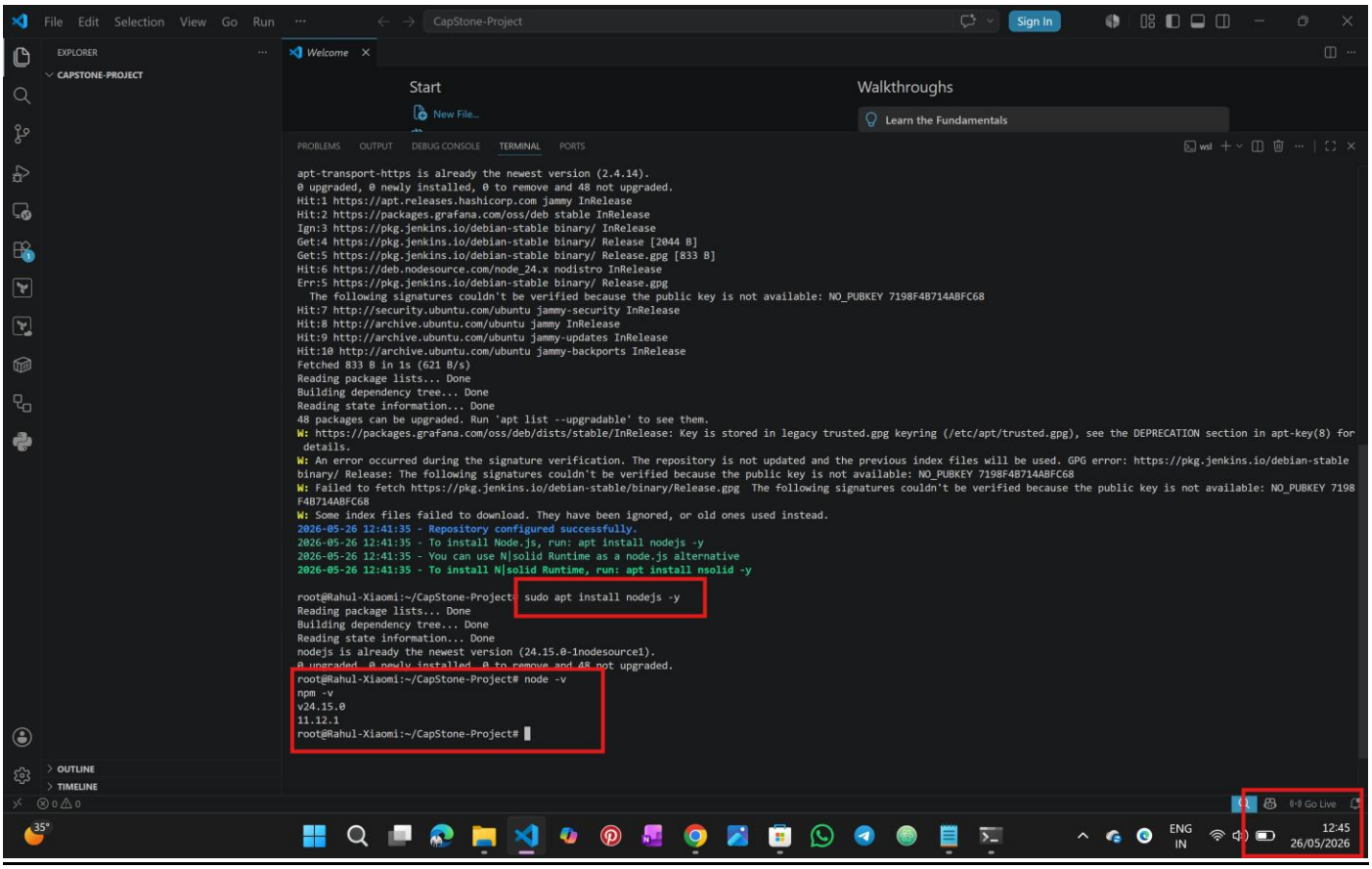
Local Environment Setup

Step 1: Install Node.js(Add NodeSource Repository For Node 24) & Install Node.js 24

curl -fsSL https://deb.nodesource.com/setup_24.x | sudo -E bash –

sudo apt install nodejs -y



Step 2: Clone Repository

Git clone git@github.com:rahulnalathati-21/AWS-3-Tier-Architecture.git

Backend Setup (Local)

Navigate to Backend Folder

cd AWS-3-Tier-Architecture/application-code/app-tier

Install Dependencies

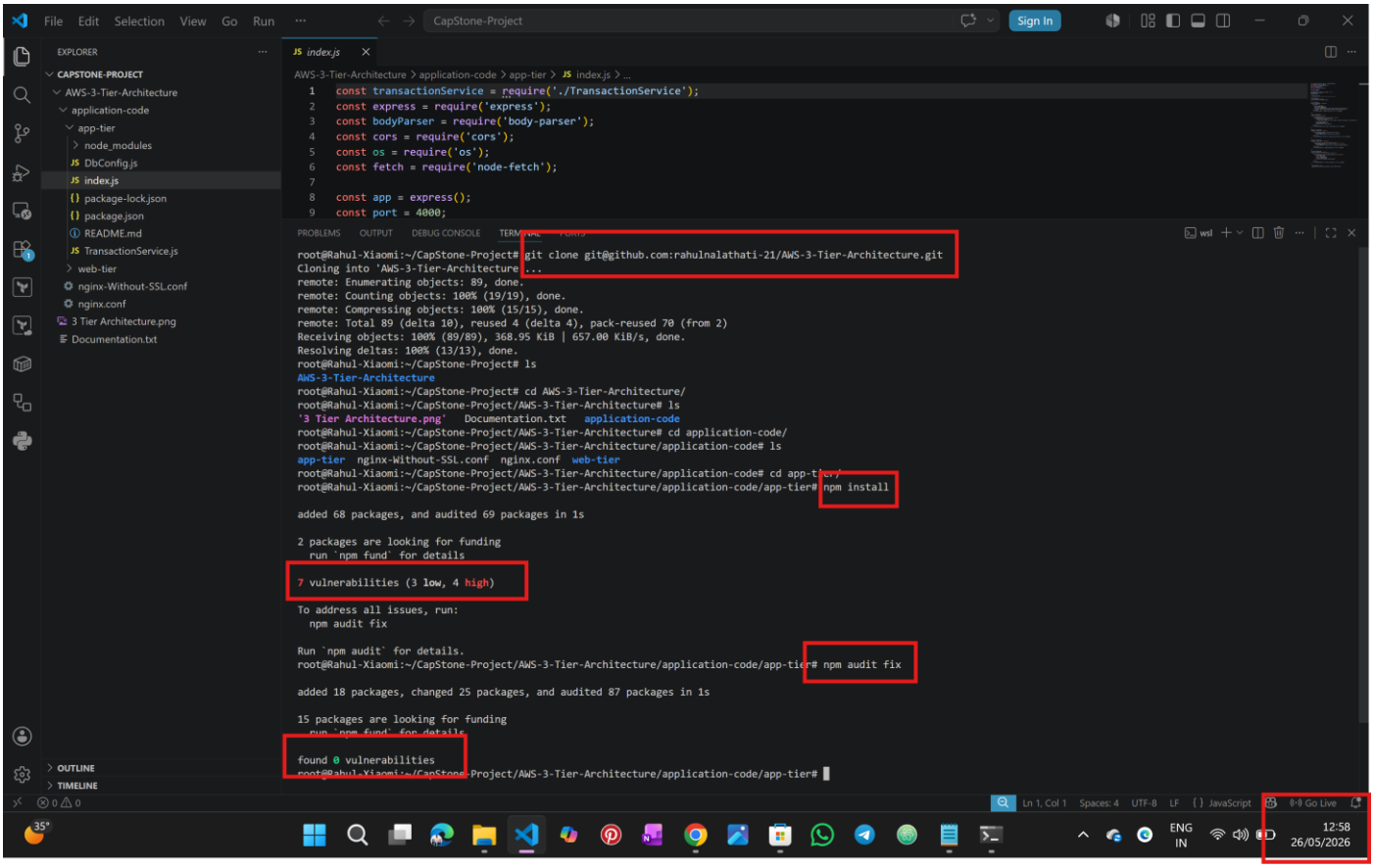
npm install

we have found there are some vulnerabilities, so we have fixed them to zero by using

npm audit fix

Run Backend

node index.js



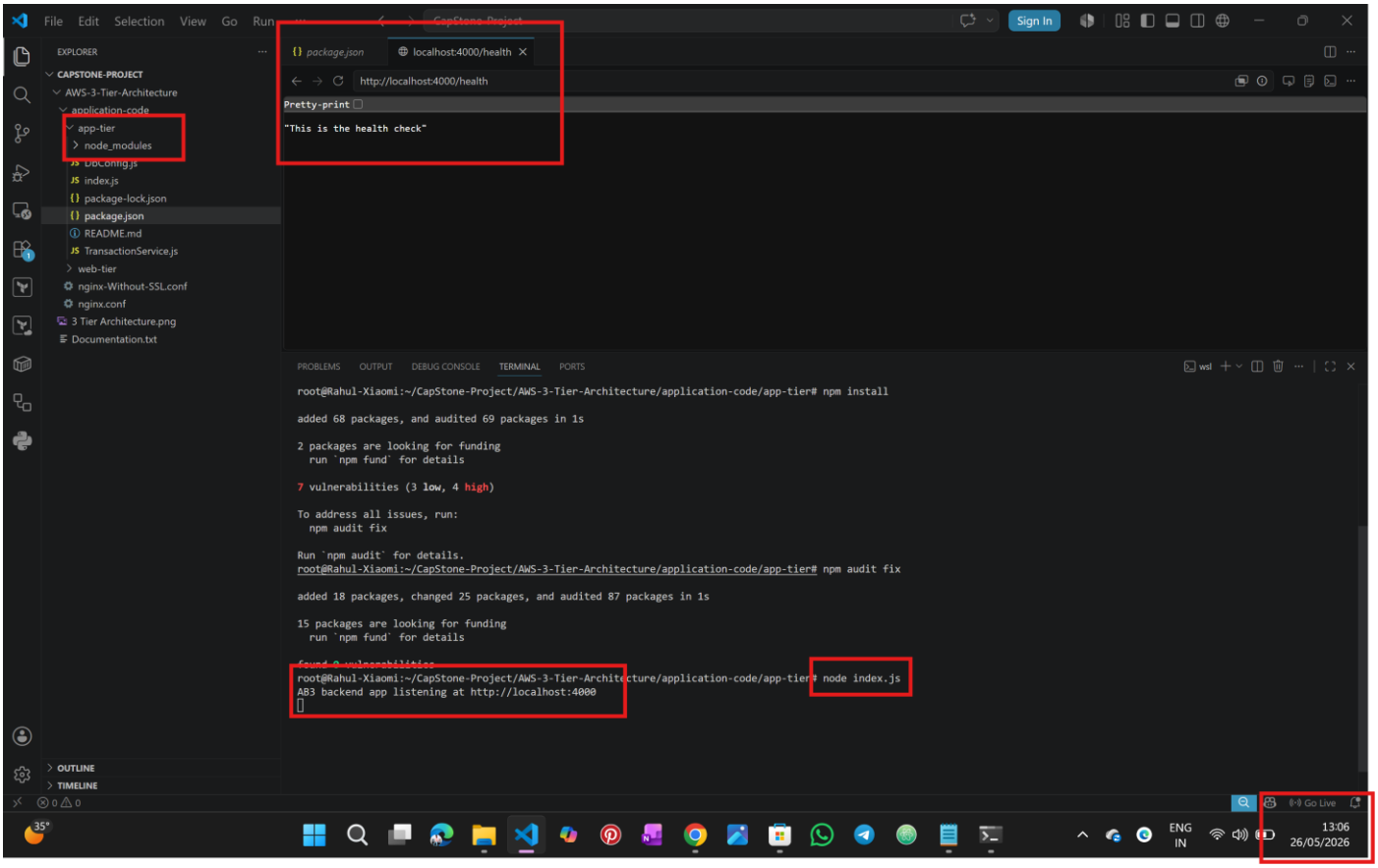
Backend Health Check

Open:

http://localhost:4000/health

Expected Output:

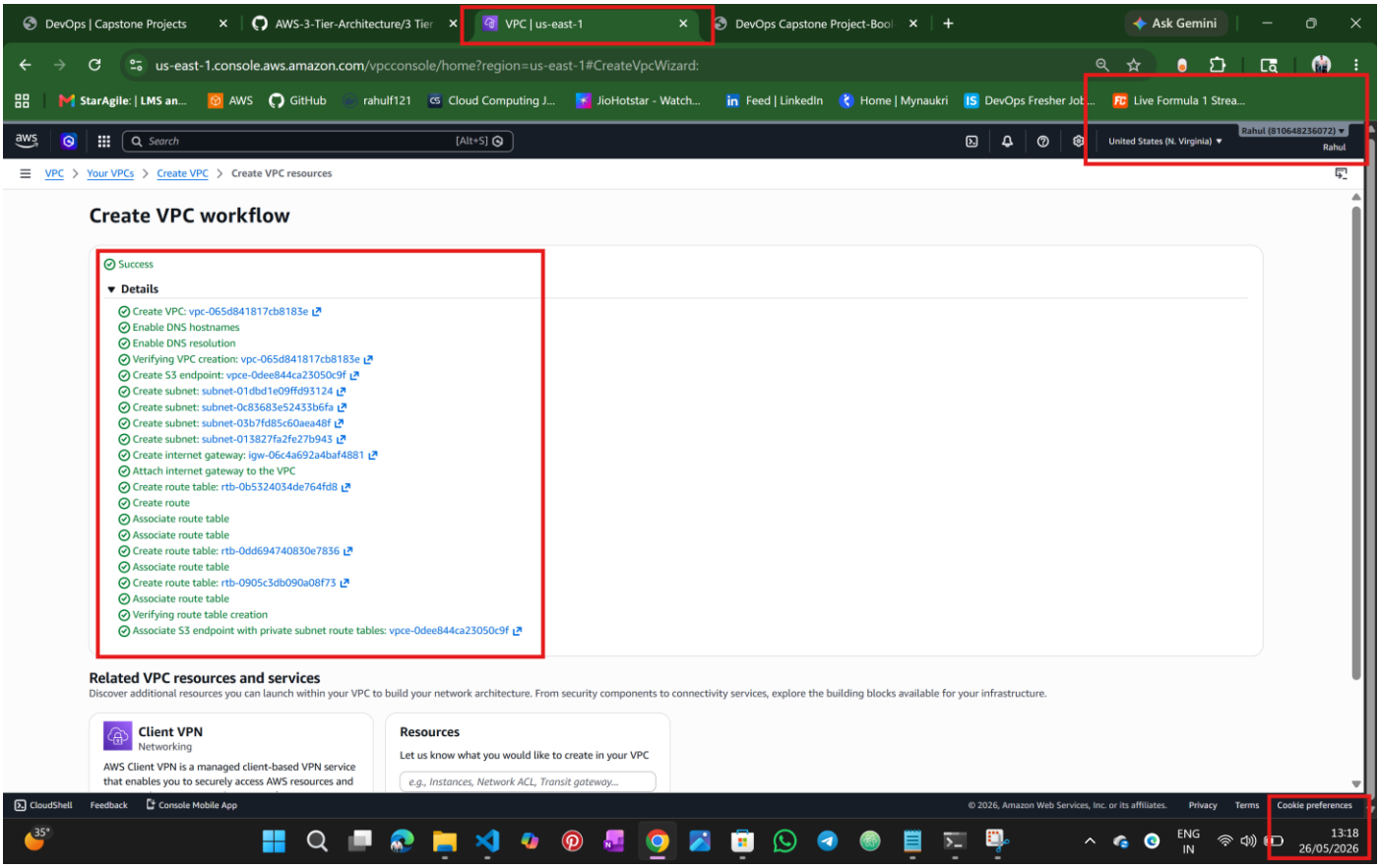
"This is the health check"



----------Local setup is working fine now I will do entire project use AWS services----------

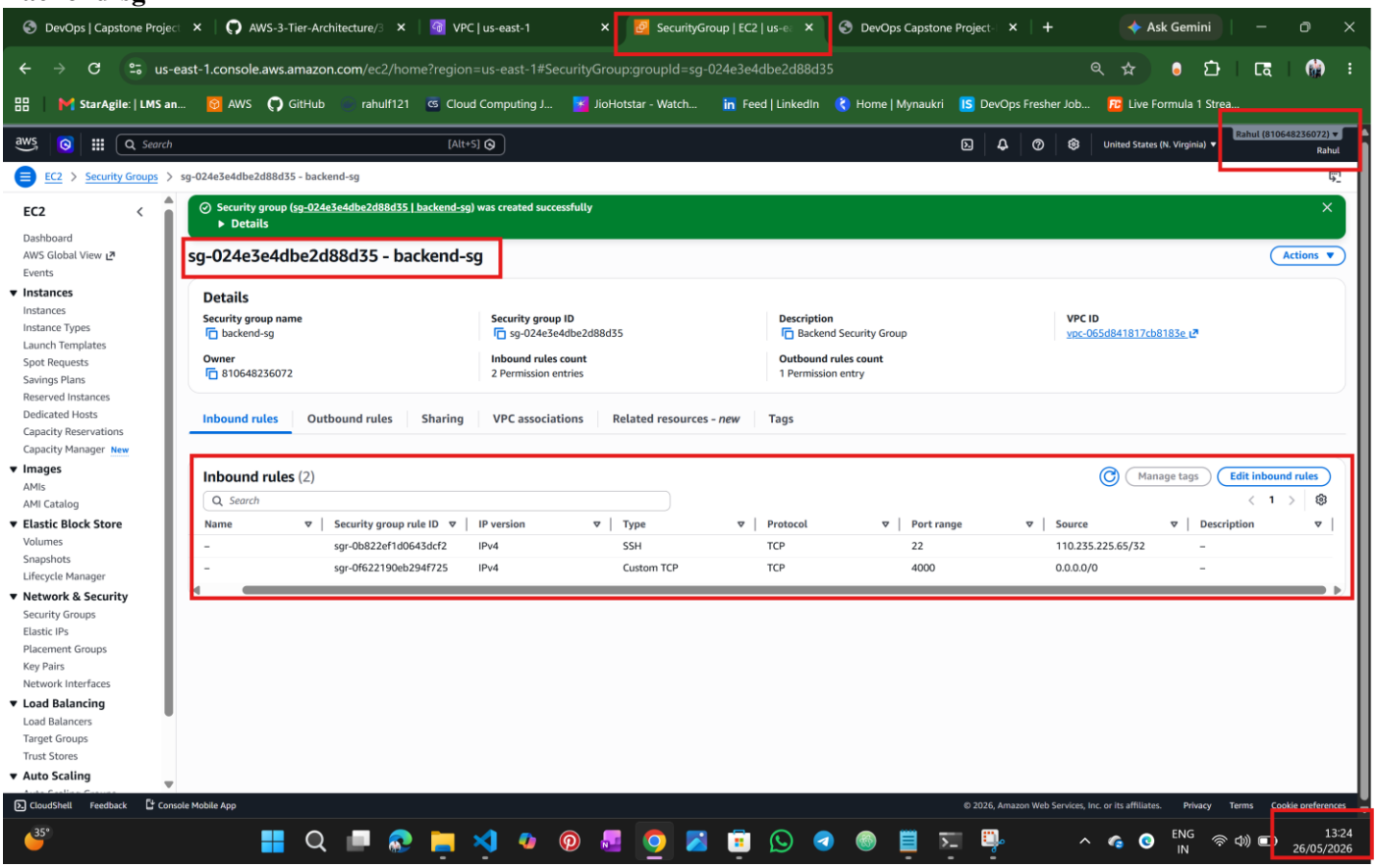
AWS Infrastructure Deployment

Login to AWS Console
Step 1: Create VPC

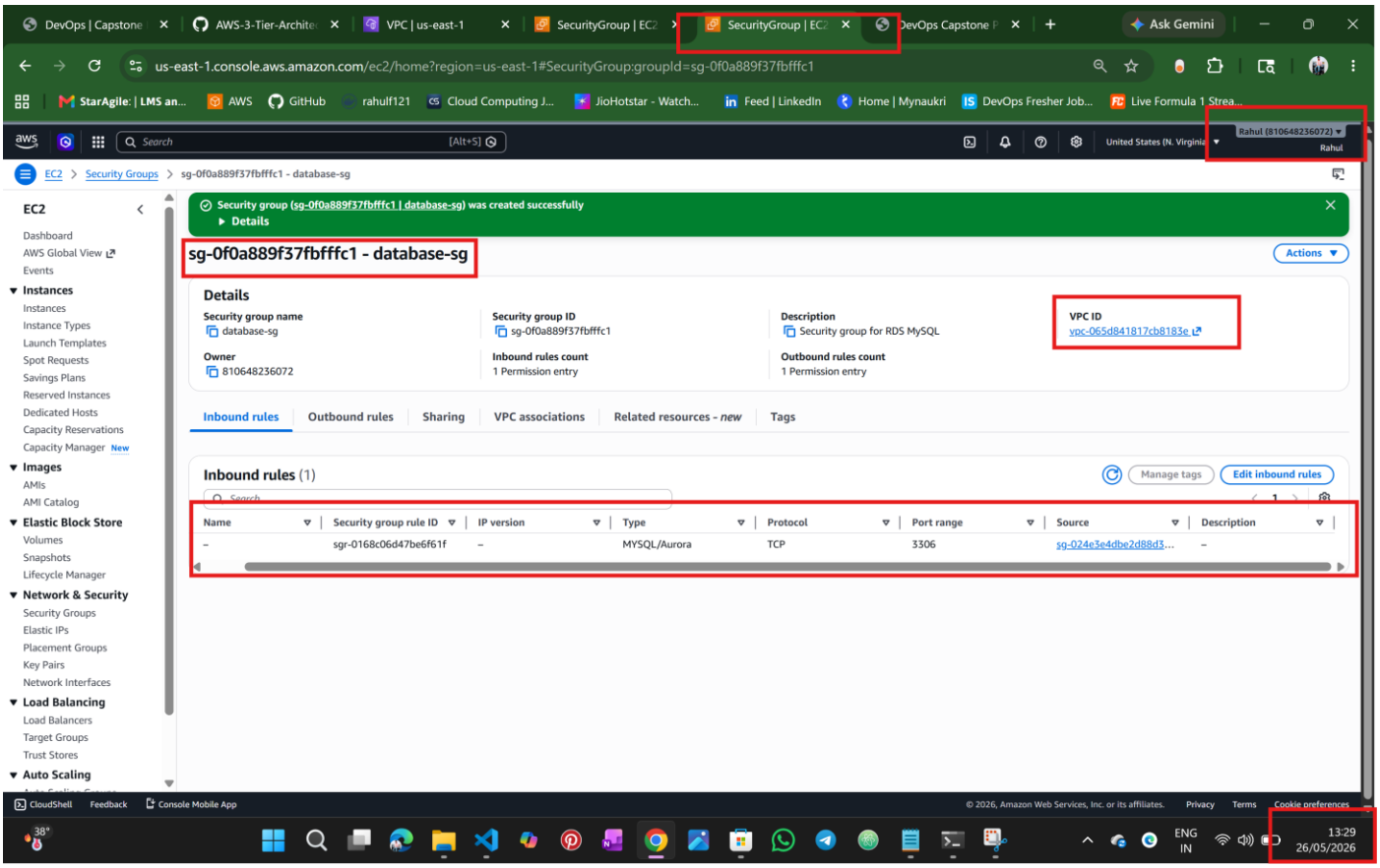


Step 2: Create Security Groups

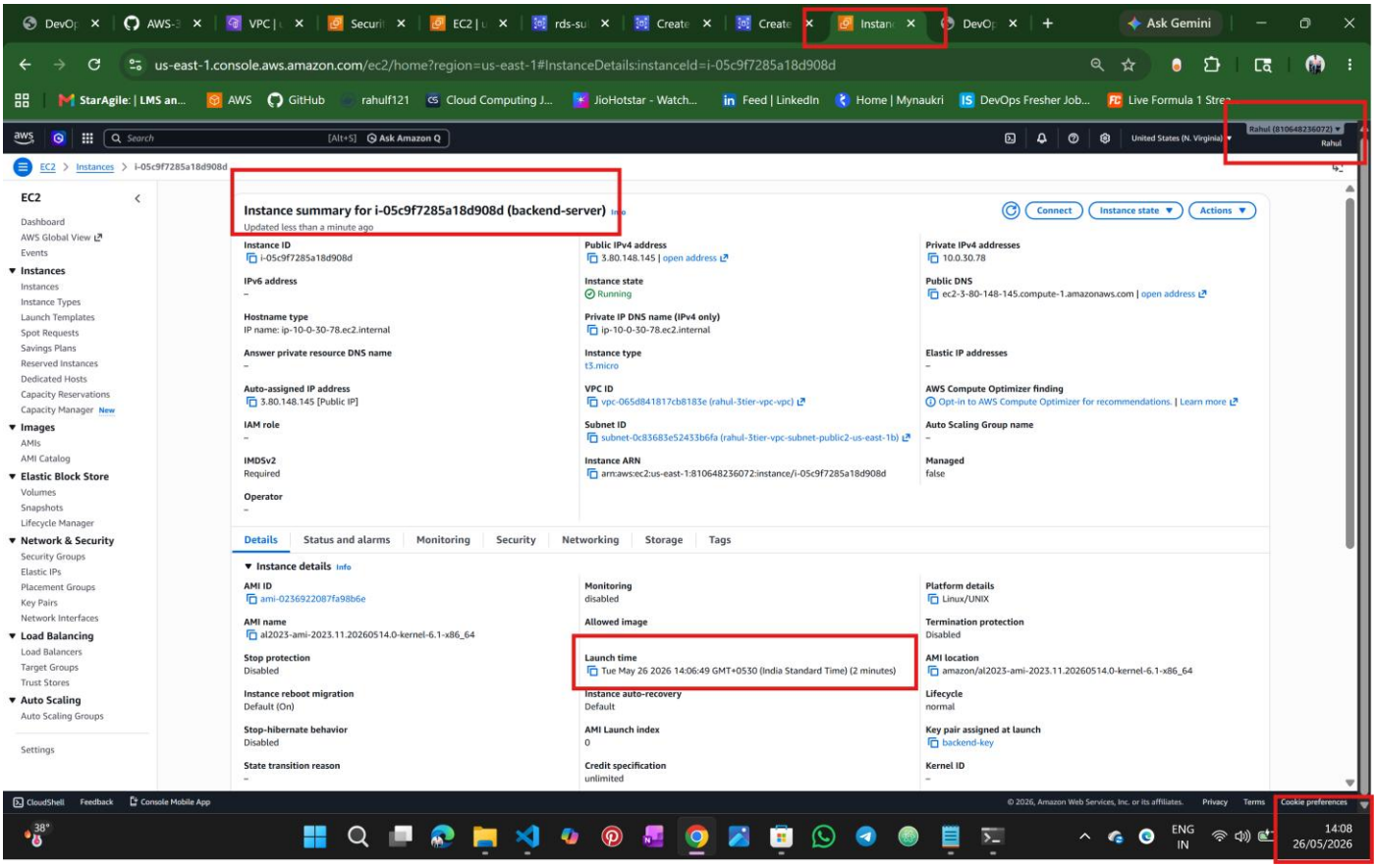
Backend-sg



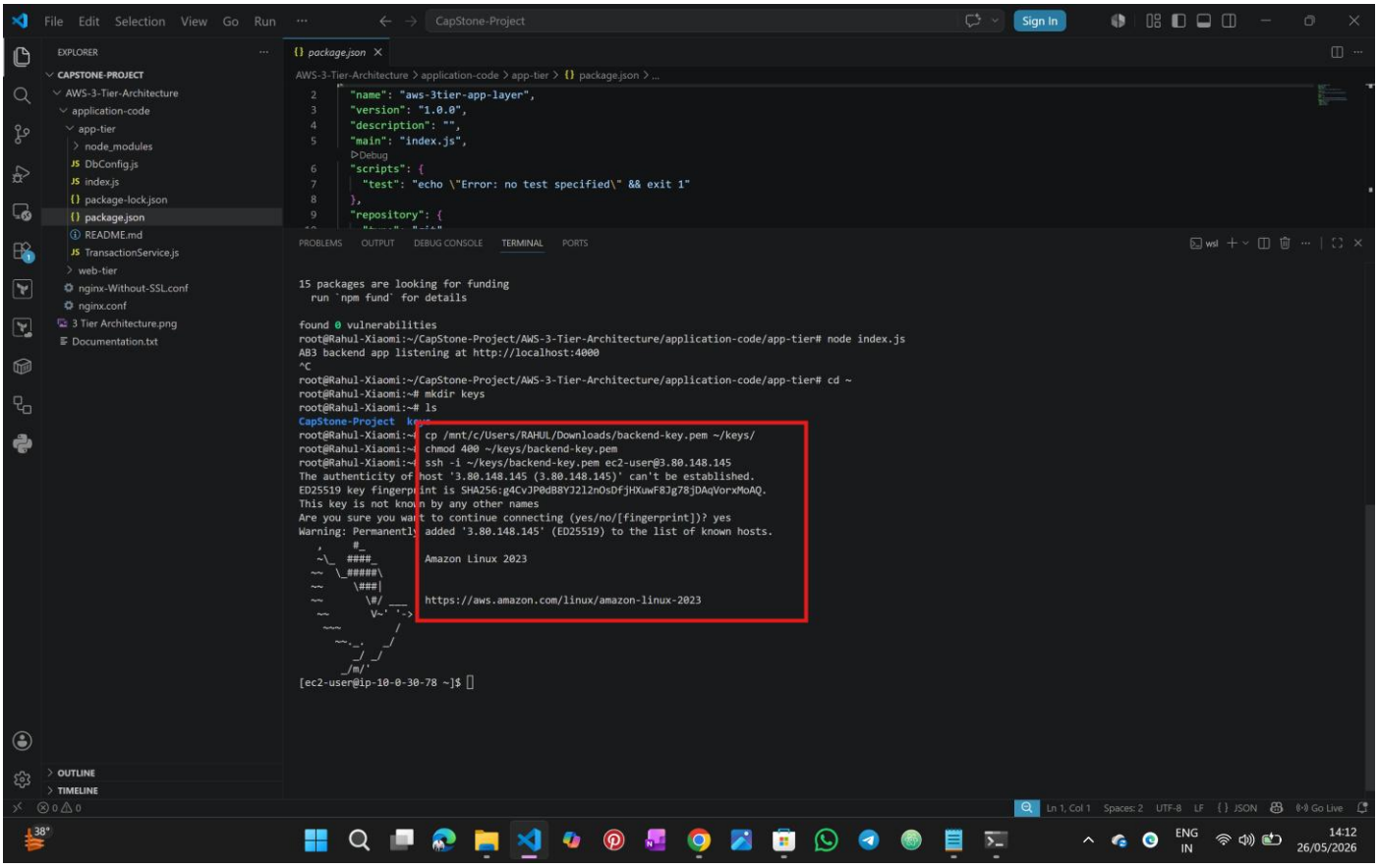
Database-SG



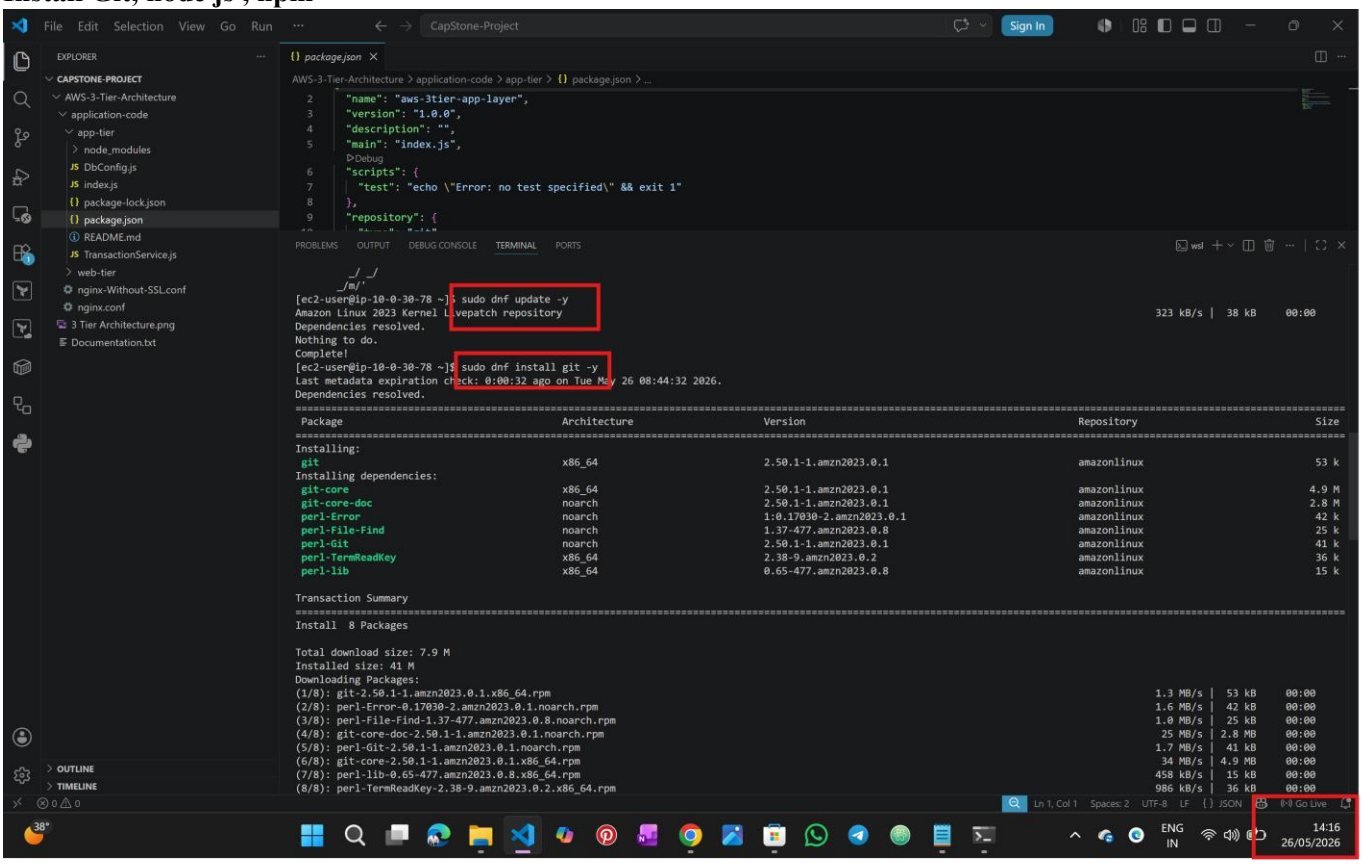
Step 3: Creating a backend server

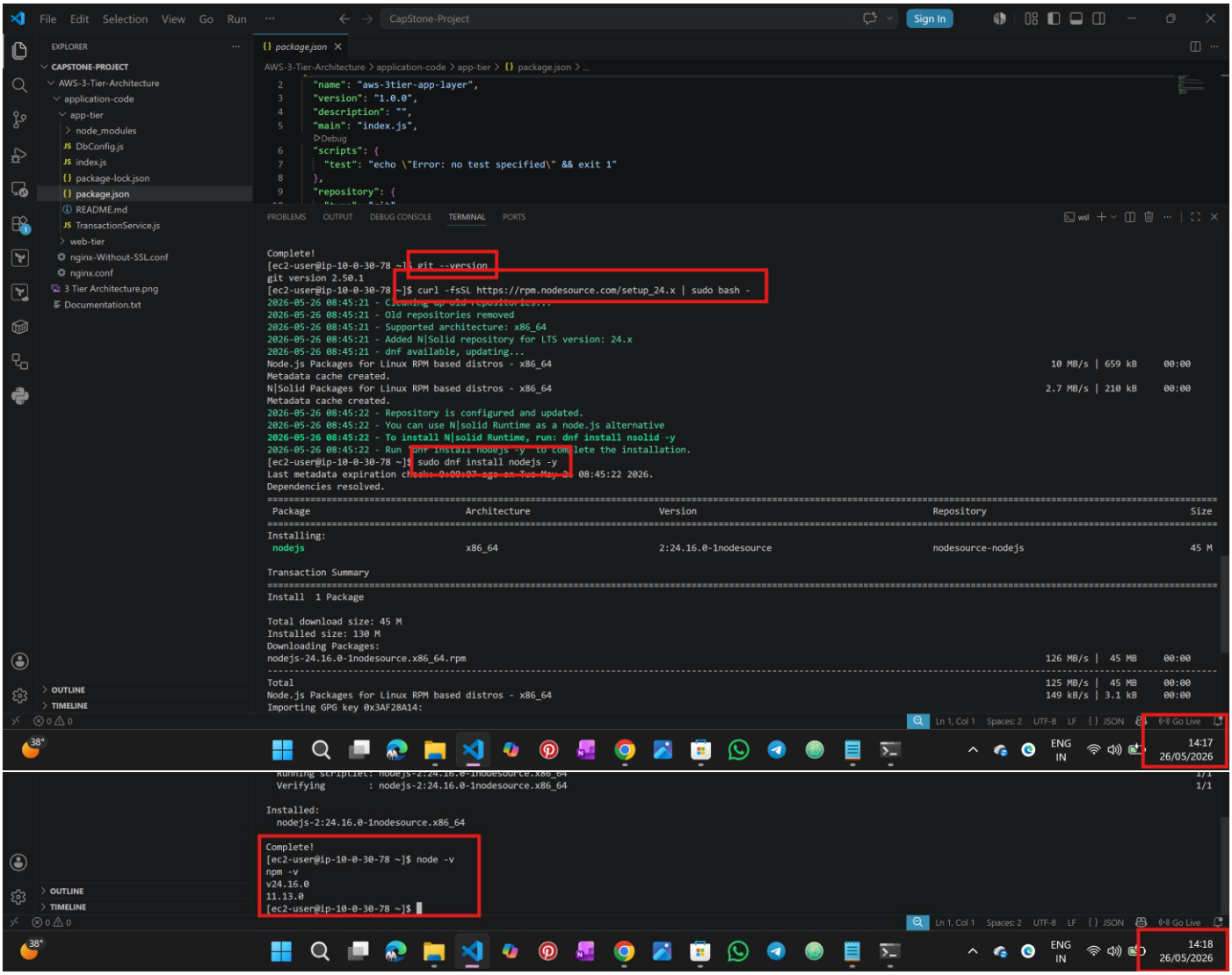


Connecting to the server

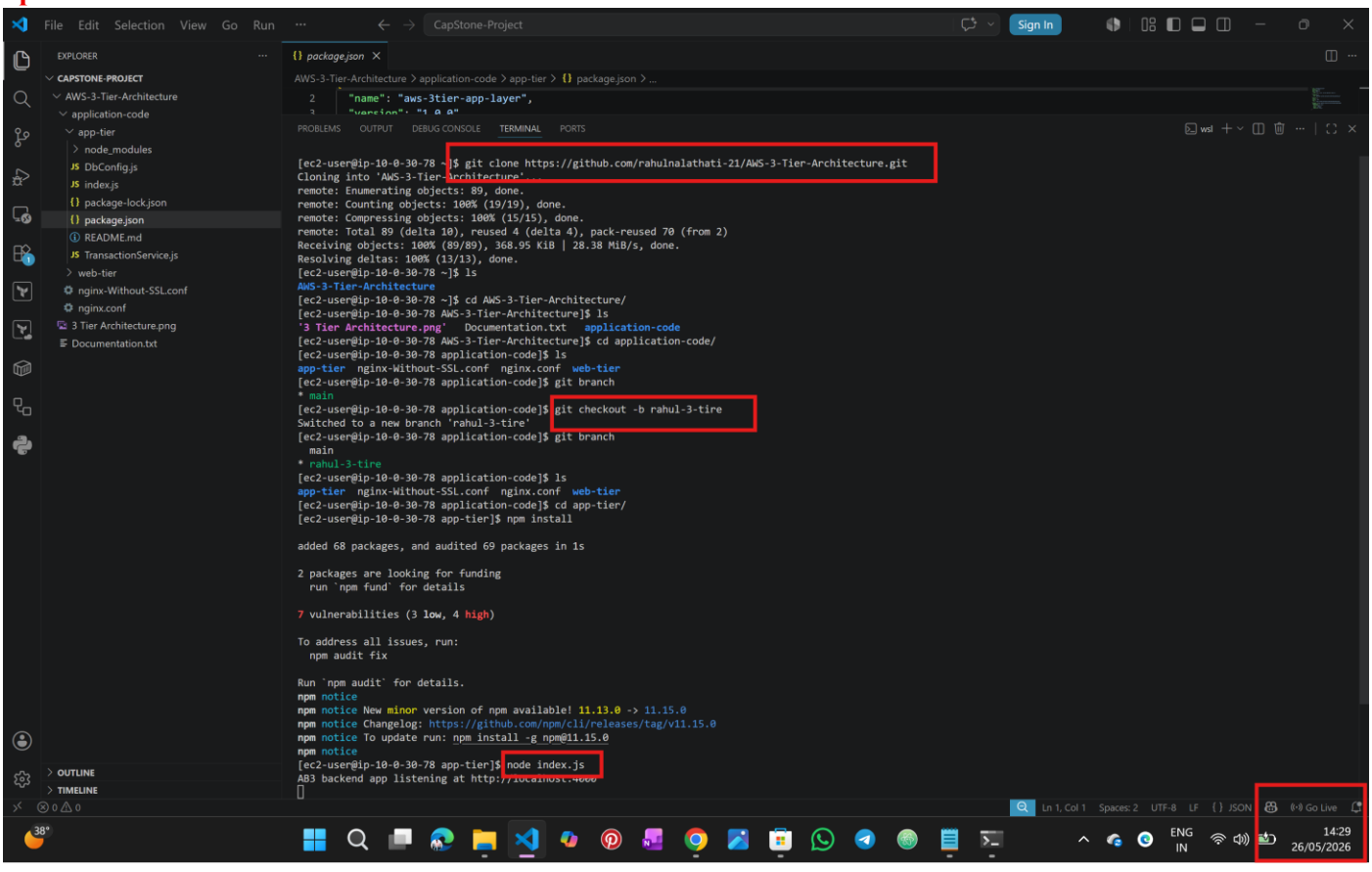


Install Git, node js , npm

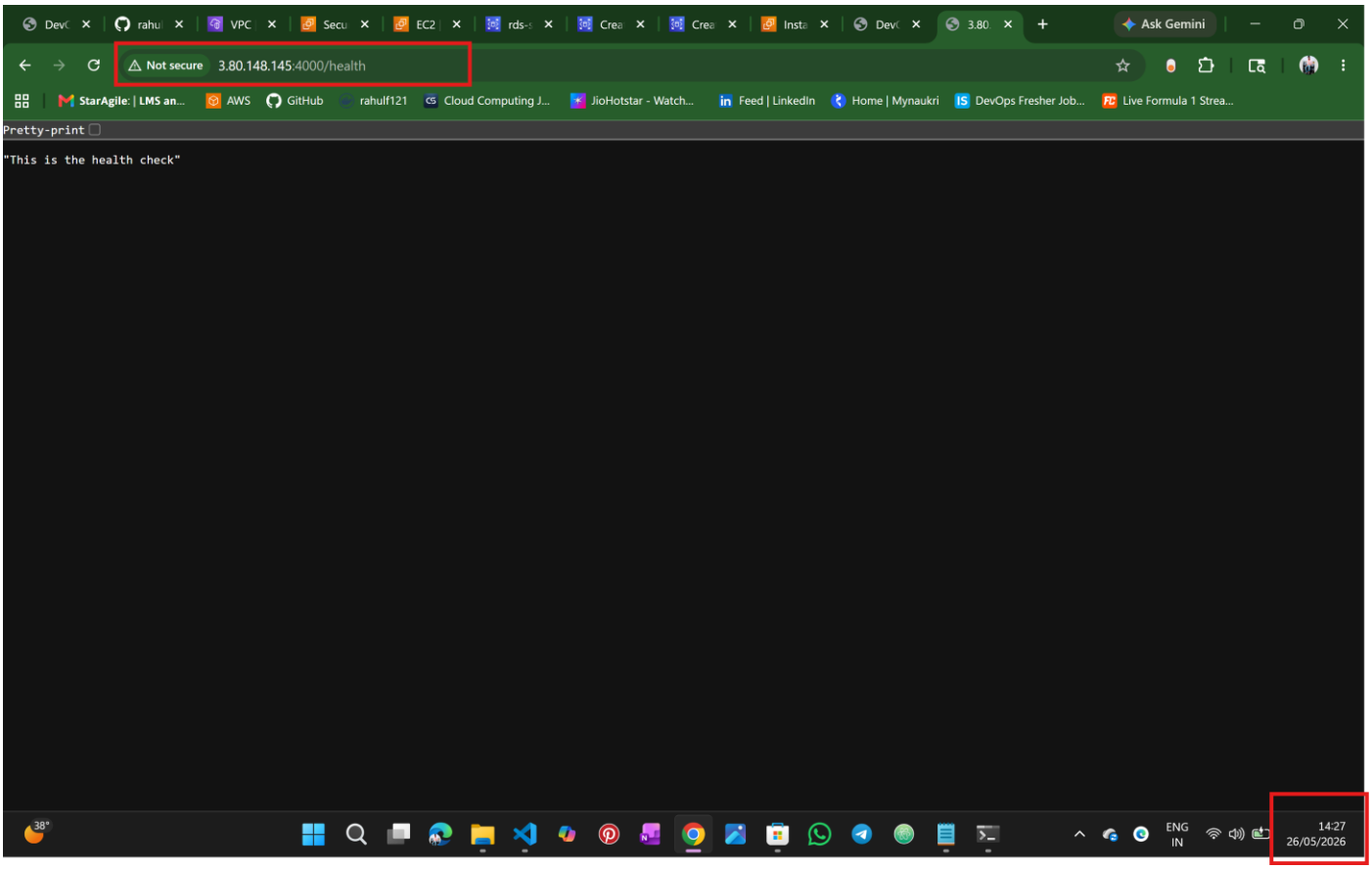




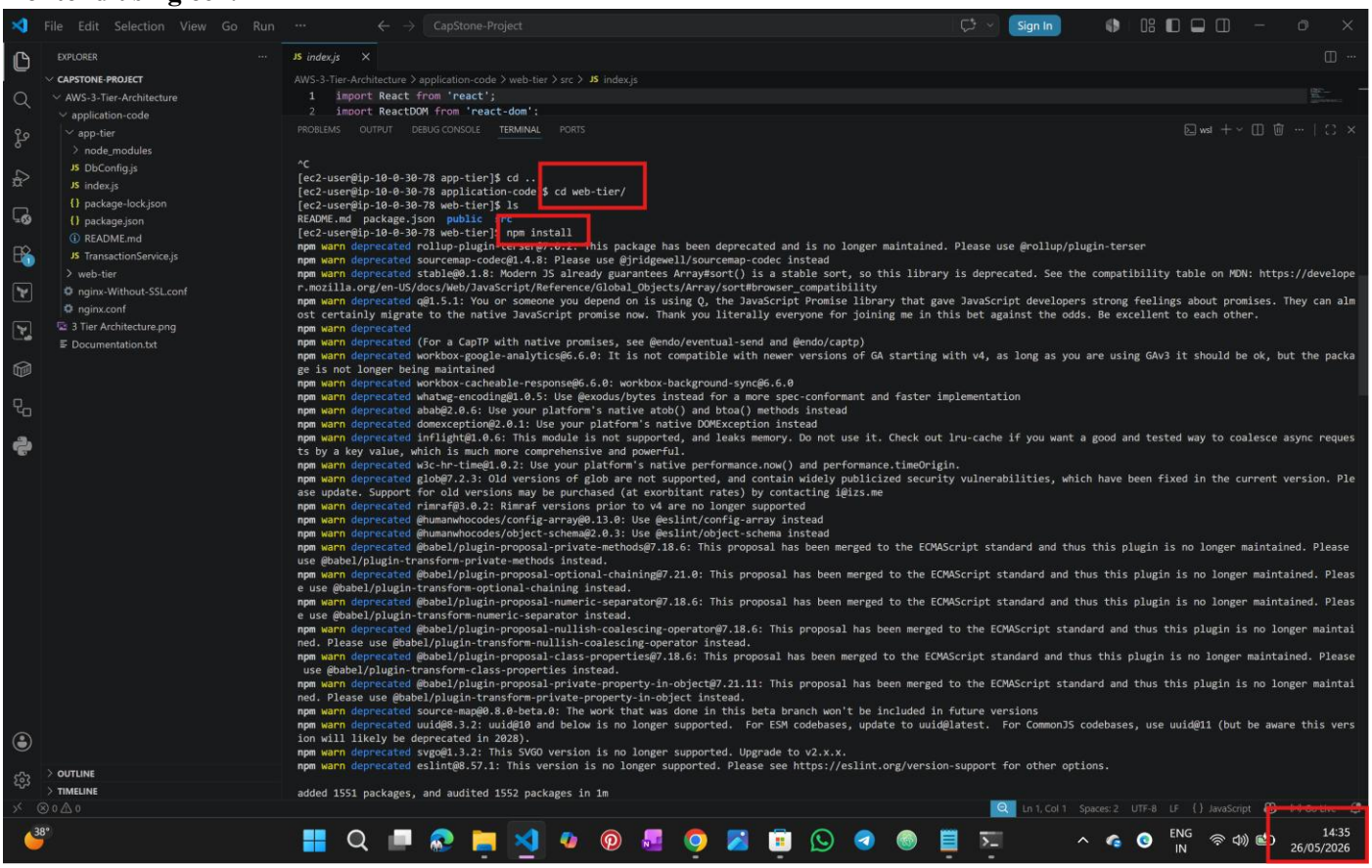
Cloned the git repo and deployed the backend I got some issues/ vulnerabilities I have fixed them using npm audit fix --force



Back end URL is working as expected



frontend using ec2:



I got some issues/ vulnerabilities I have fixed them using
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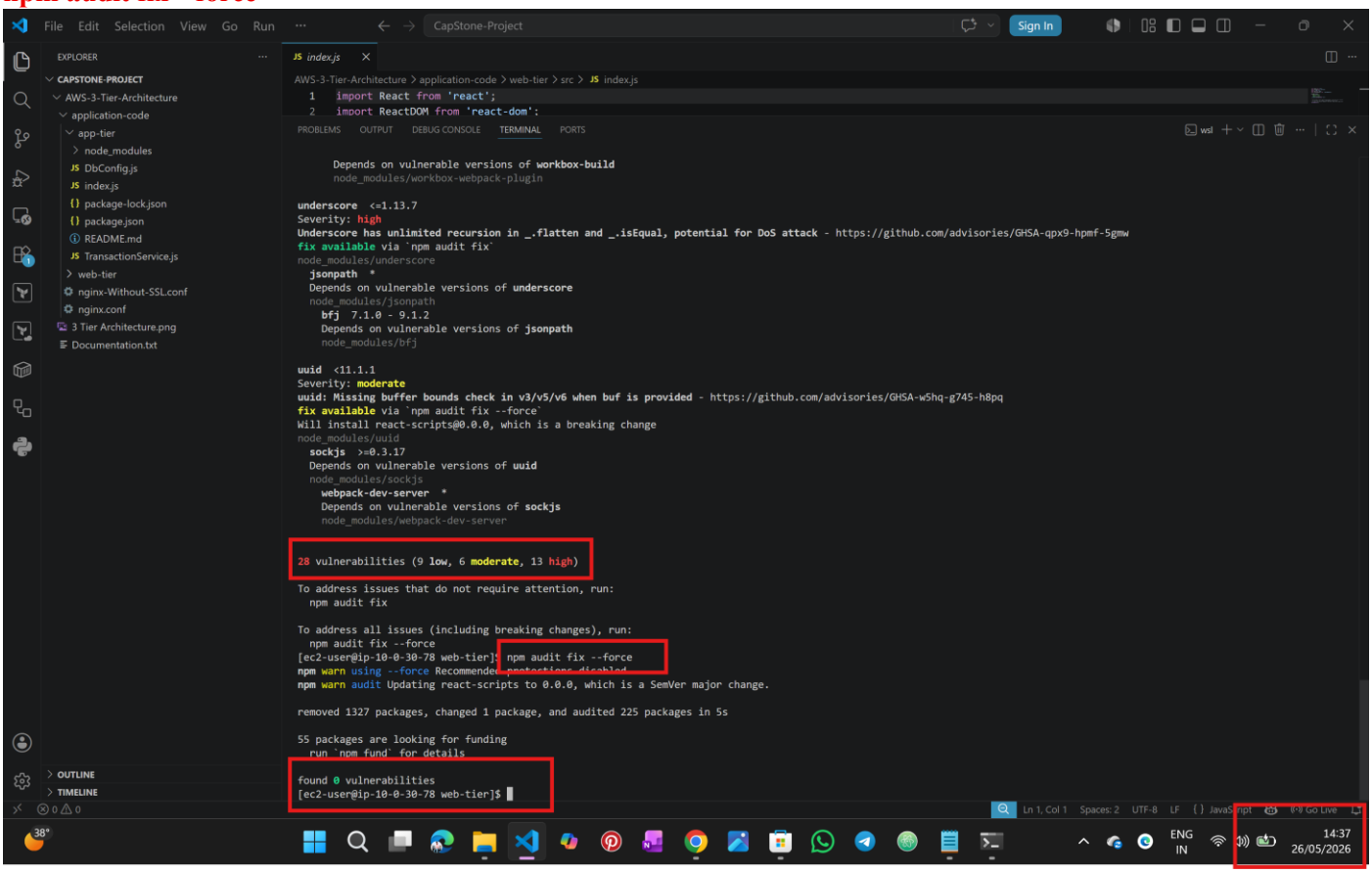
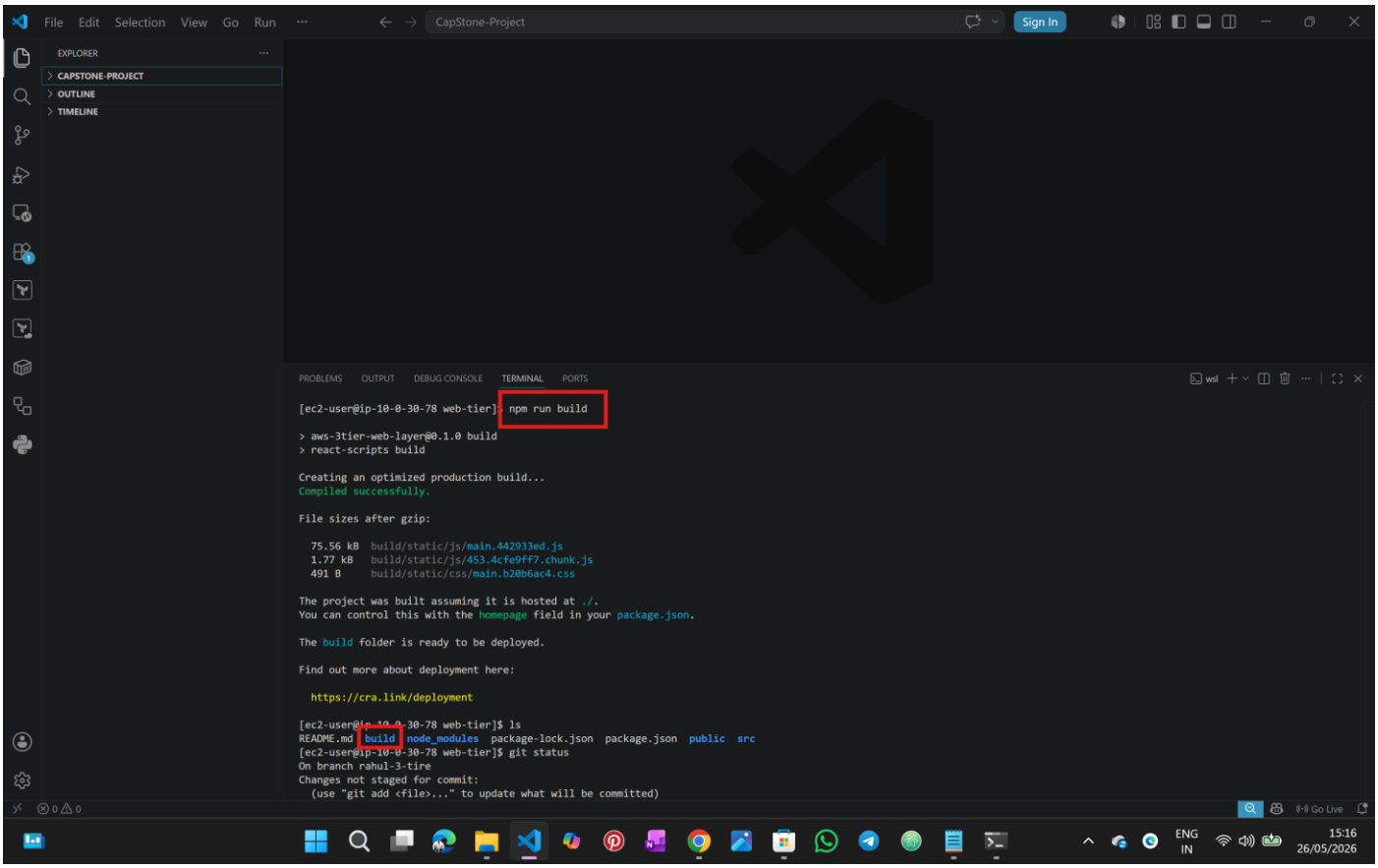
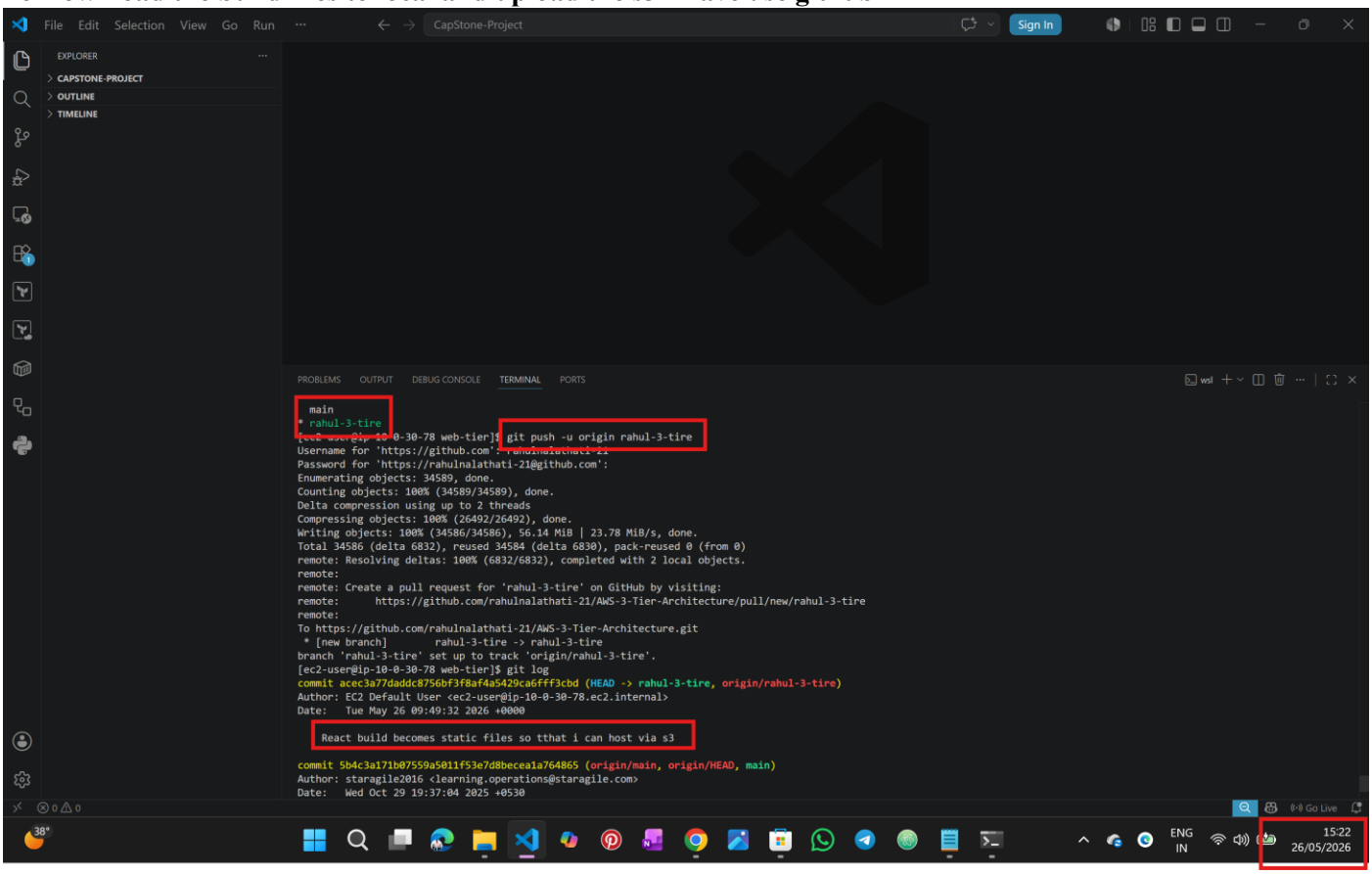
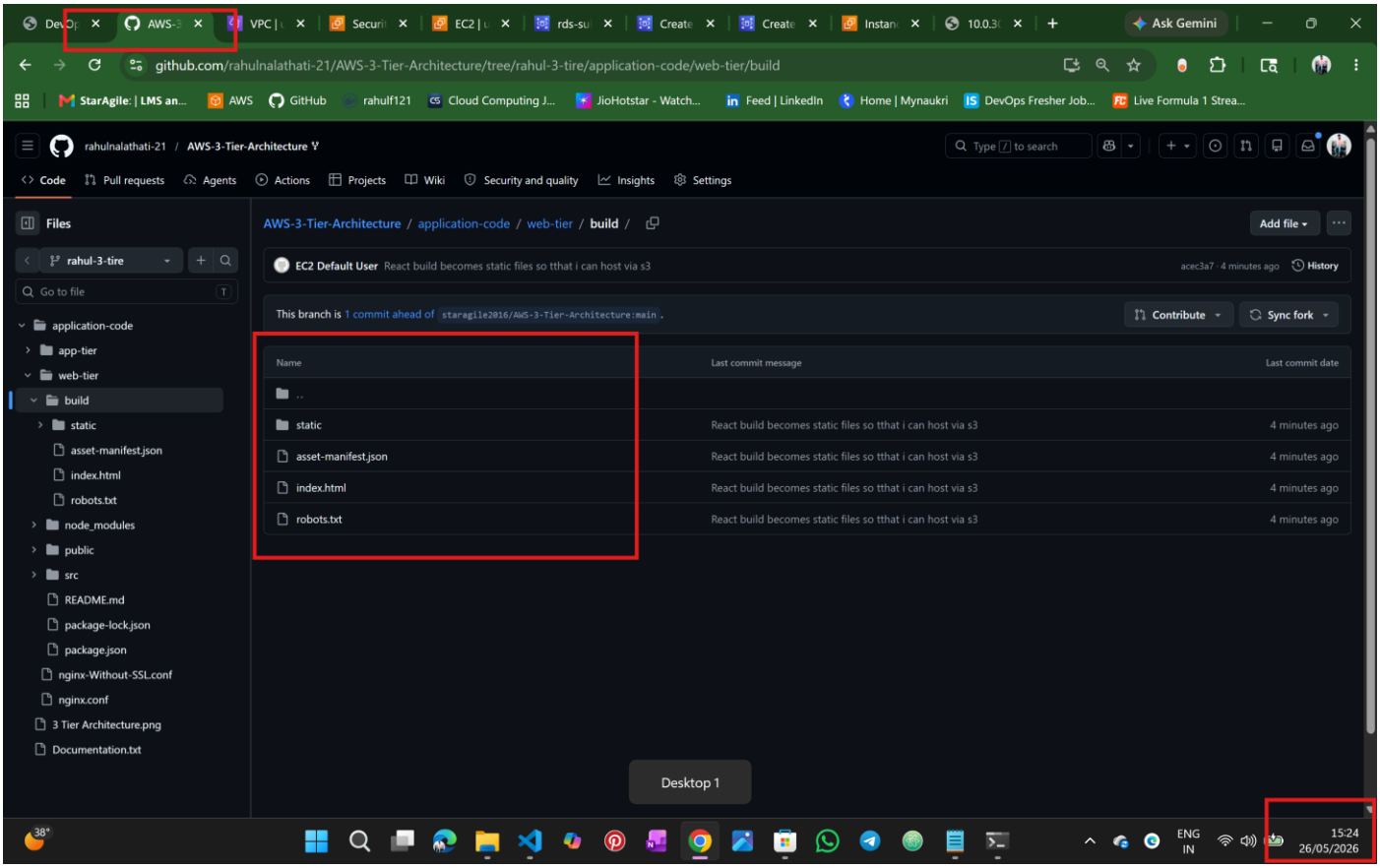


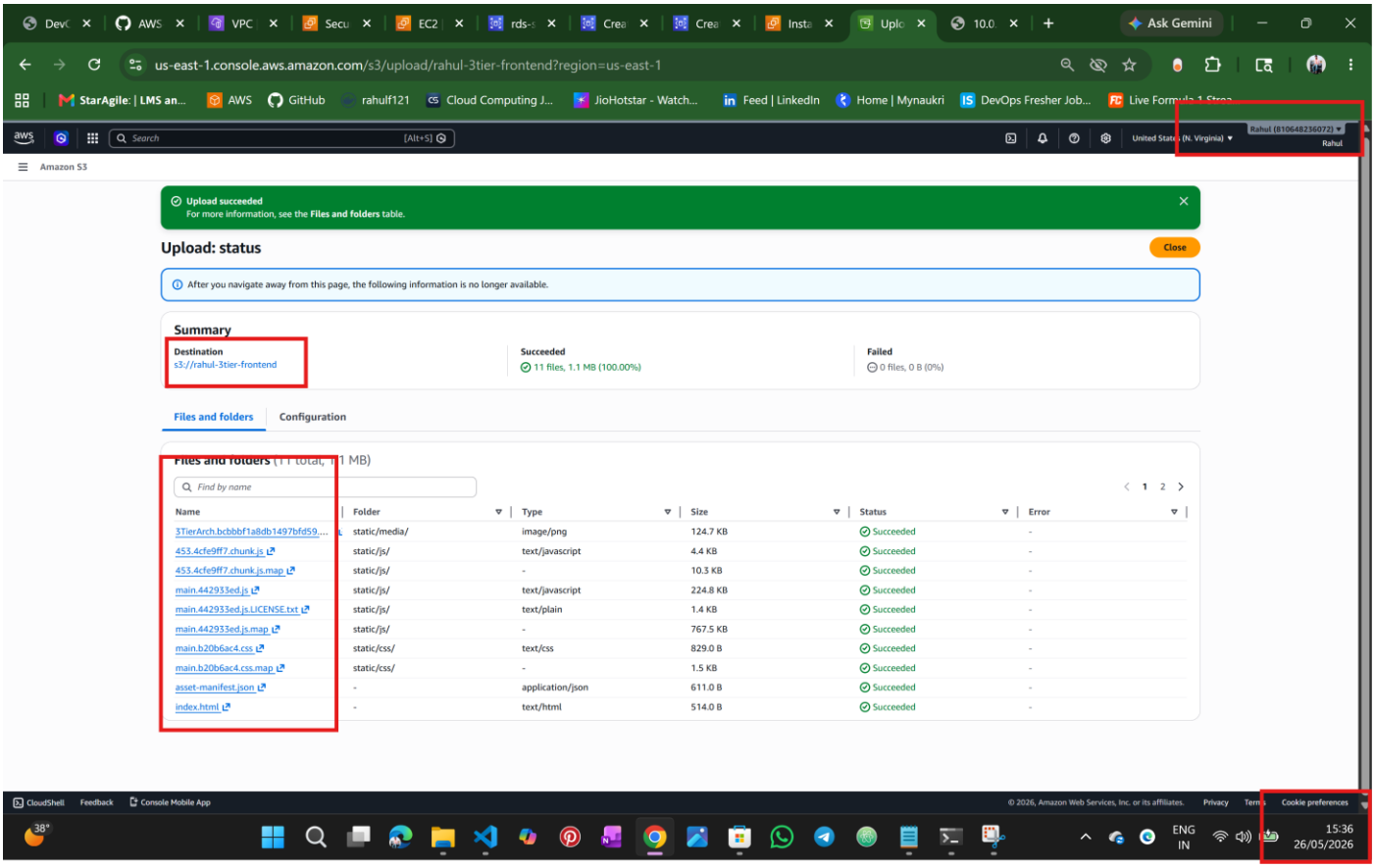
Since we are using the static web hosting we need to build and upload because React build becomes static files.

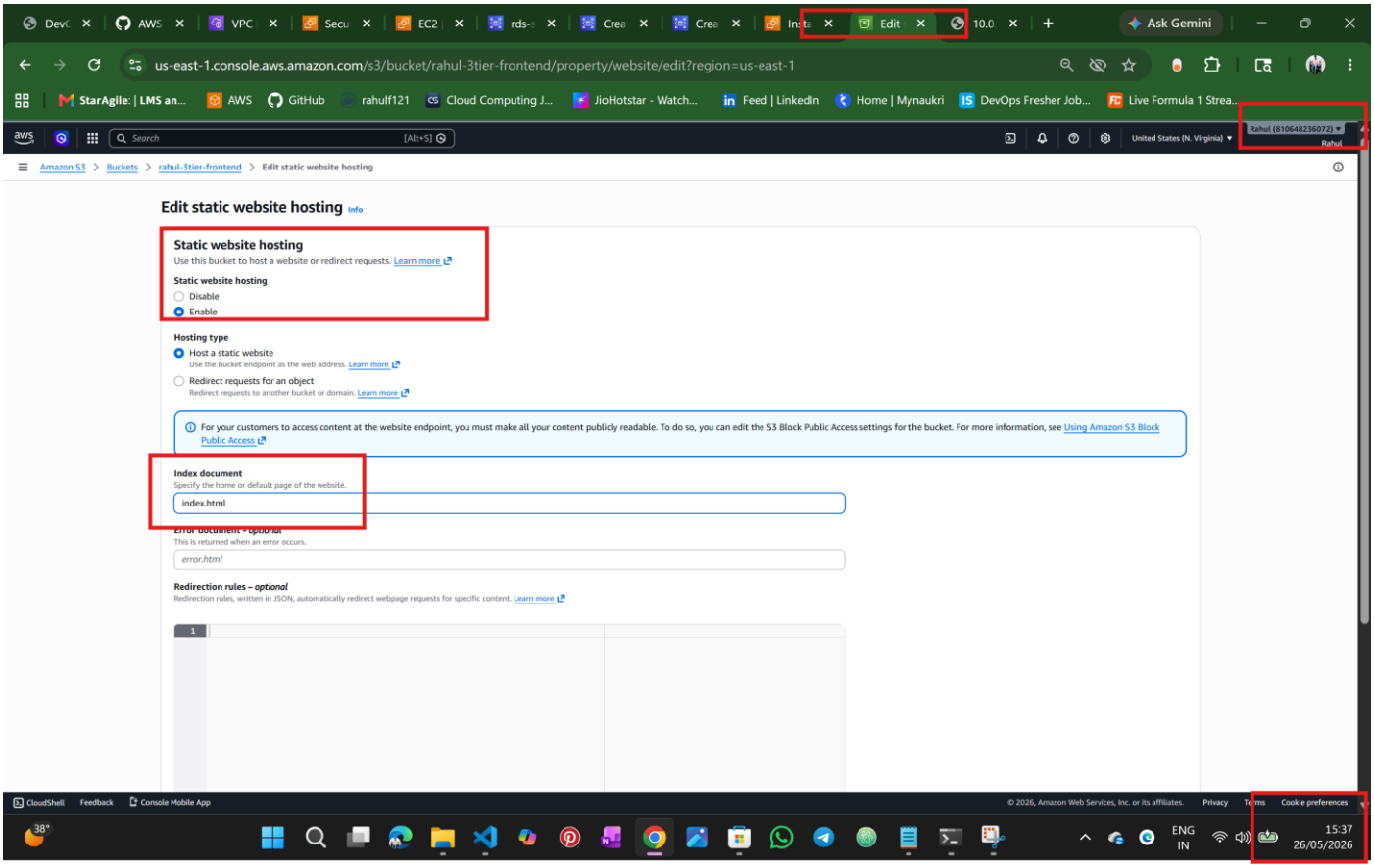


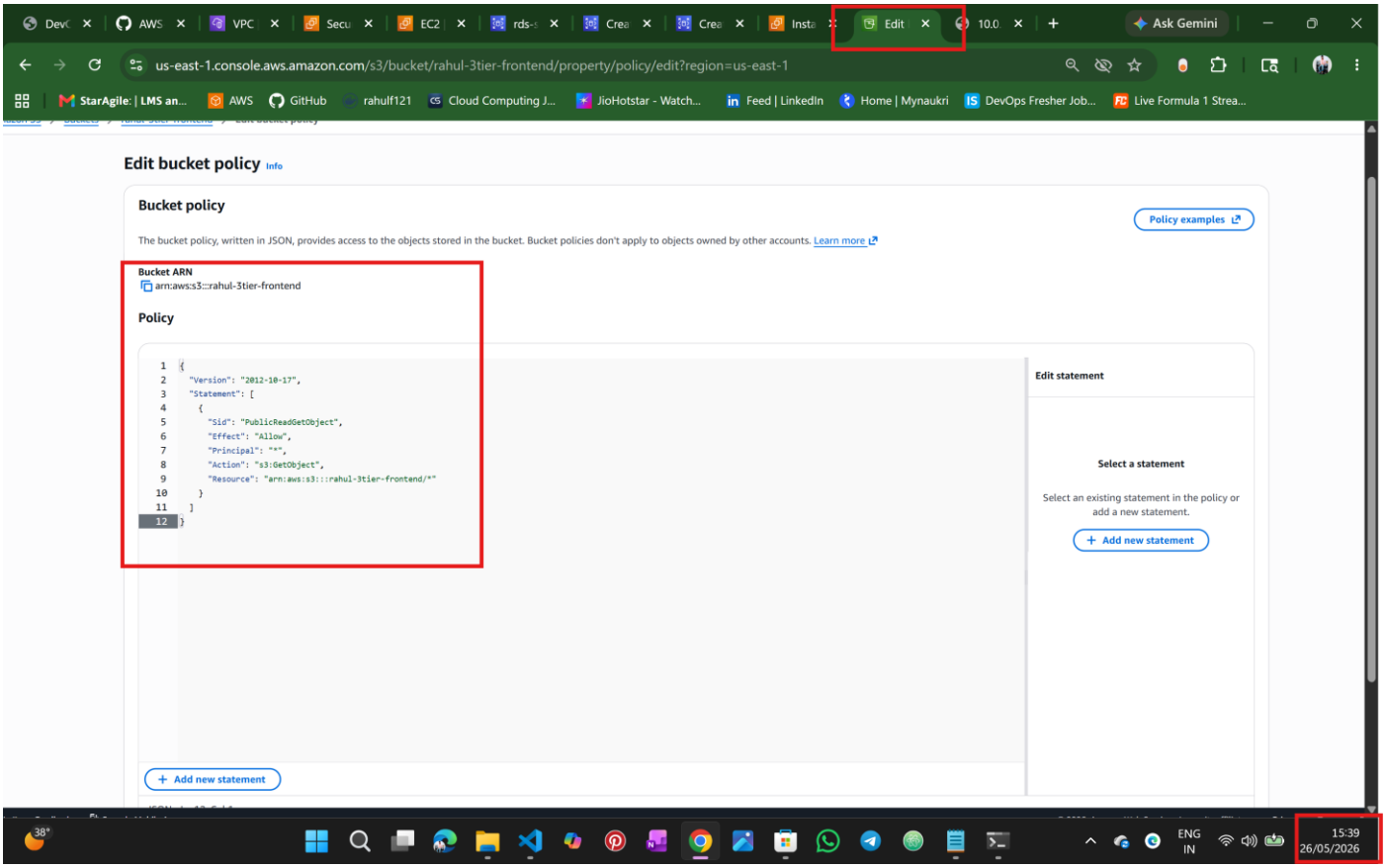
To Download the build files to local and upload the s3 I have use github

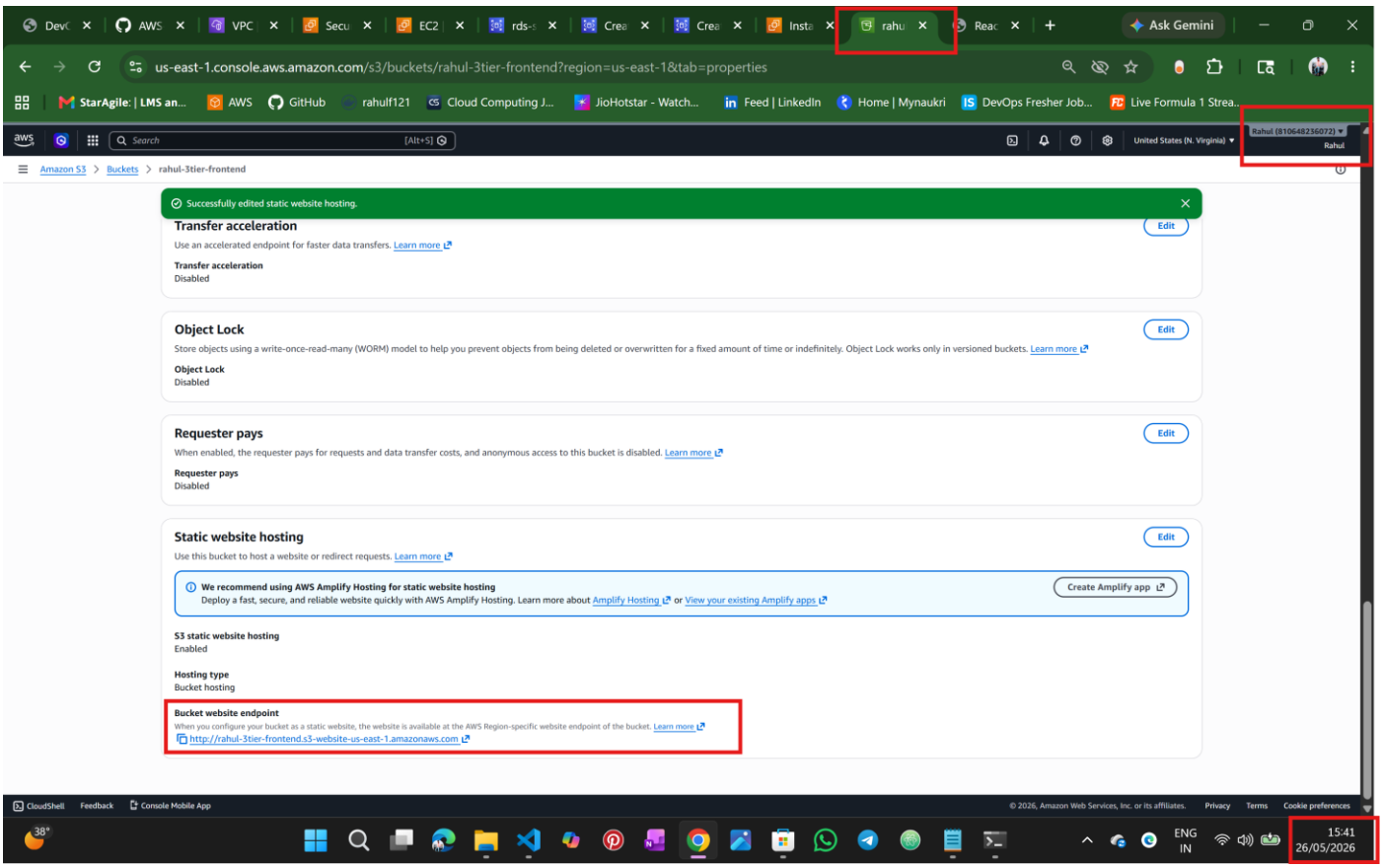


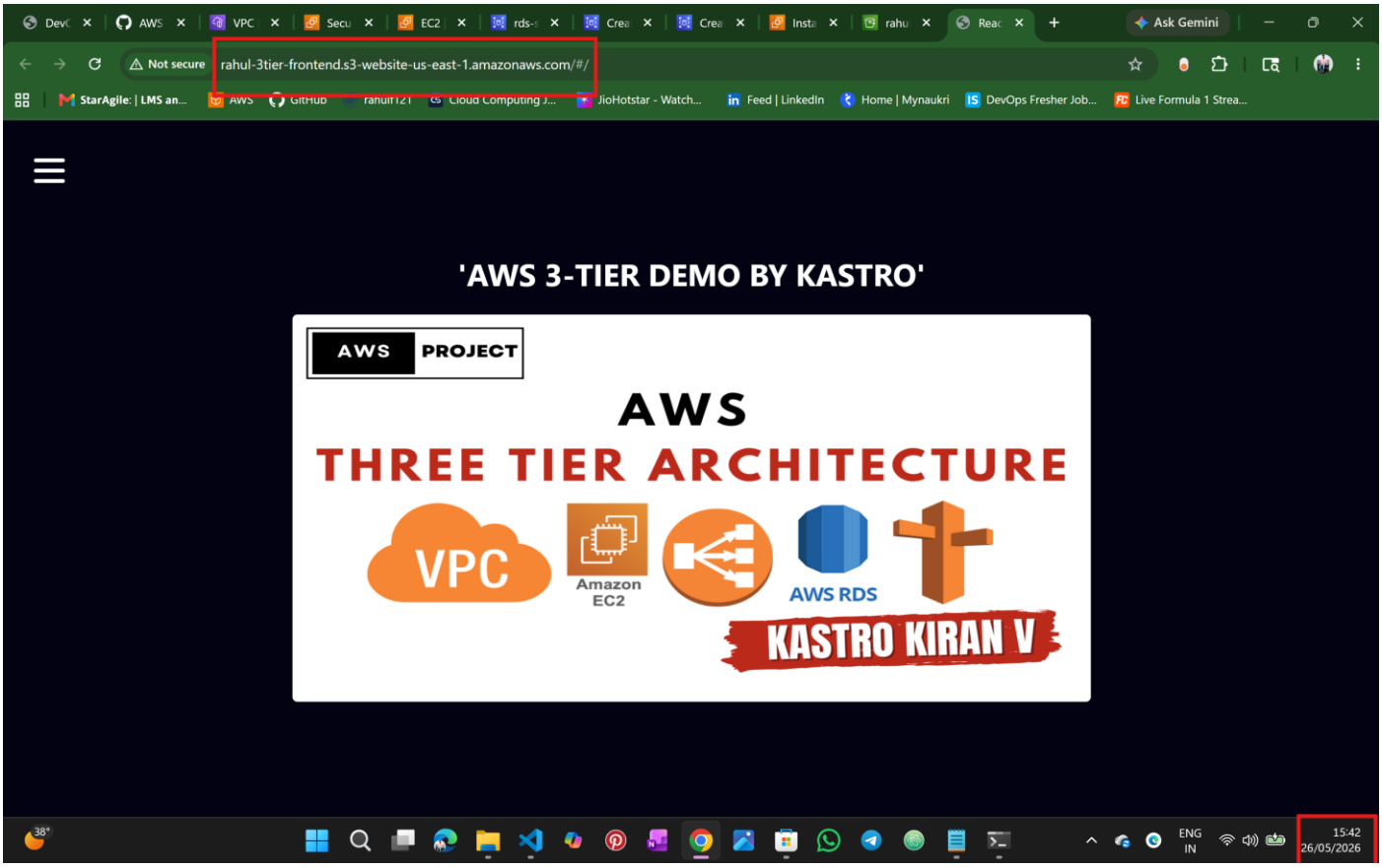



us-east-1.console.aws.amazon.com/s3/upload/rahul-3tier-frontend?region=us-east-1
Upload succeeded
For more information, see the Files and folders table.
Upload: status
Close
After you navigate away from this page, the following information is no longer available.
Summary
Destination
s3://rahul-3tier-frontend
Succeeded
11 files, 1.1 MB (100.00%)
Failed
0 files, 0 B (0%)
Files and folders
Configuration
Files and folders (11 total, 1.1 MB)
Find by name
Name
Folder
Type
Size
Status
Error
3TierArch.bcbbbf1a8db1497bfd59....
static/media/
image/png
124.7 KB
Succeeded
453.4cfe9ff7.chunk.js
static/js/
text/javascript
4.4 KB
Succeeded
453.4cfe9ff7.chunk.js.map
static/js/
10.3 KB
Succeeded
main.442933ed.js
static/js/
text/javascript
224.8 KB
Succeeded
main.442933ed.js.LICENSE.txt
static/js/
text/plain
1.4 KB
Succeeded
main.442933ed.js.map
static/js/
767.5 KB
Succeeded
main.b20b6ac4.css
static/css/
text/css
829.0 B
Succeeded
main.b20b6ac4.css.map
static/css/
1.5 KB
Succeeded
asset-manifest.json
application/json
611.0 B
Succeeded
index.html
text/html
514.0 B
Succeeded


us-east-1.console.aws.amazon.com/s3/bucket/rahul-3tier-frontend/property/website/edit?region=us-east-1
Amazon S3 > Buckets > rahul-3tier-frontend > Edit static website hosting
Edit static website hosting
Info
Static website hosting
Use this bucket to host a website or redirect requests. Learn more
Static website hosting
Disable
Enable
Hosting type
Host a static website
Use the bucket endpoint as the web address. Learn more
Redirect requests for an object
Redirect requests to another bucket or domain. Learn more
For your customers to access content at the website endpoint, you must make all your content publicly readable. To do so, you can edit the S3 Block Public Access settings for the bucket. For more information, see Using Amazon S3 Block Public Access
Index document
Specify the home or default page of the website.
index.html
Error document - optional
This is returned when an error occurs.
error.html
Redirection rules – optional
Redirection rules, written in JSON, automatically redirect webpage requests for specific content. Learn more

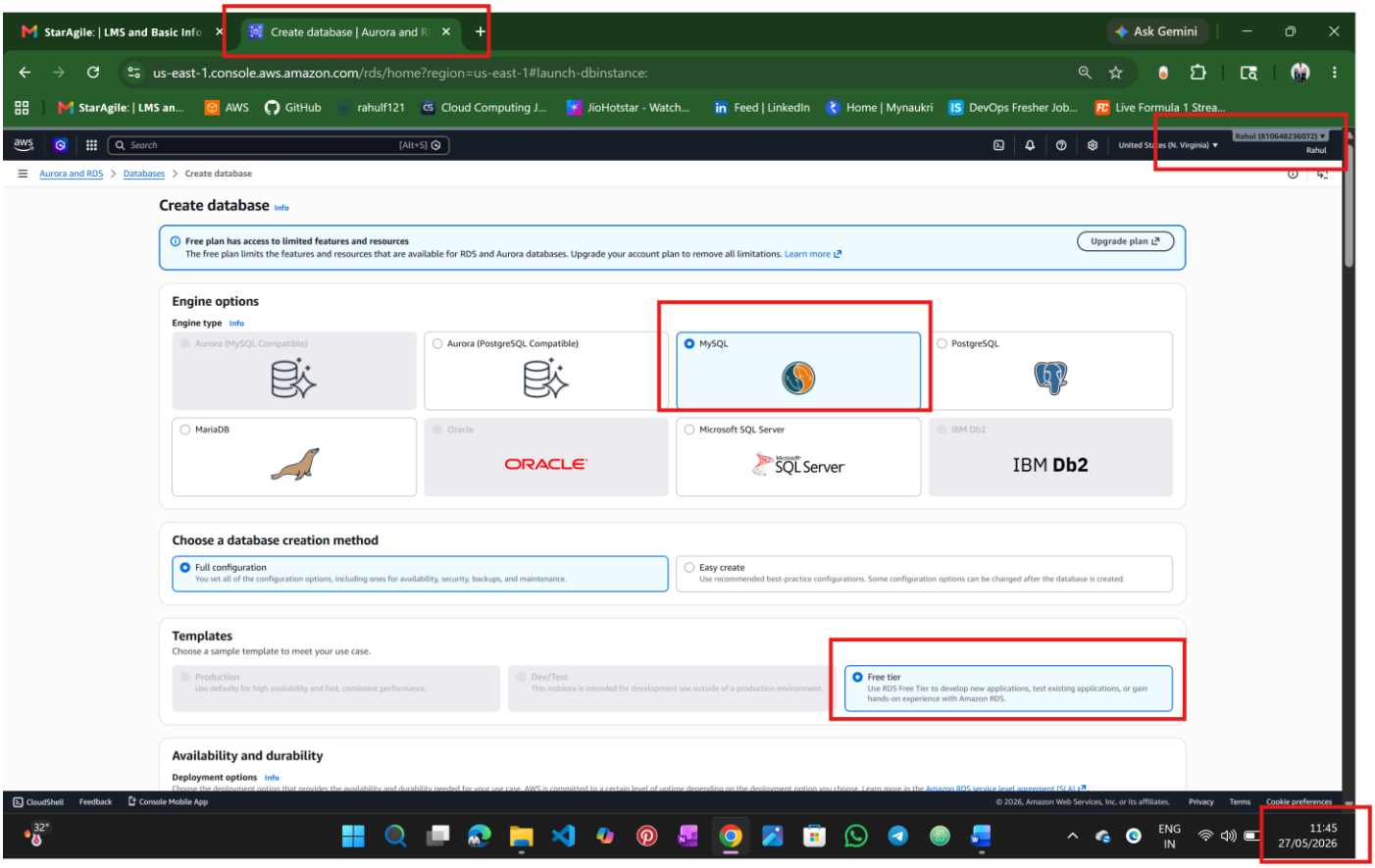

us-east-1.console.aws.amazon.com/s3/bucket/rahul-3tier-frontend/property/policy/edit?region=us-east-1
Edit bucket policy Info
Bucket policy
Policy examples
The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. Learn more
Bucket ARN
arn:aws:s3:::rahul-3tier-frontend
Policy
{
"Version": "2012-10-17",
"Statement": [
{
"Sid": "PublicReadGetObject",
"Effect": "Allow",
"Principal": "*",
"Action": "s3:GetObject",
"Resource": "arn:aws:s3:::rahul-3tier-frontend/*"
}
]
}
Edit statement
Select a statement
Select an existing statement in the policy or add a new statement.
+ Add new statement
+ Add new statement
15:39
26/05/2026

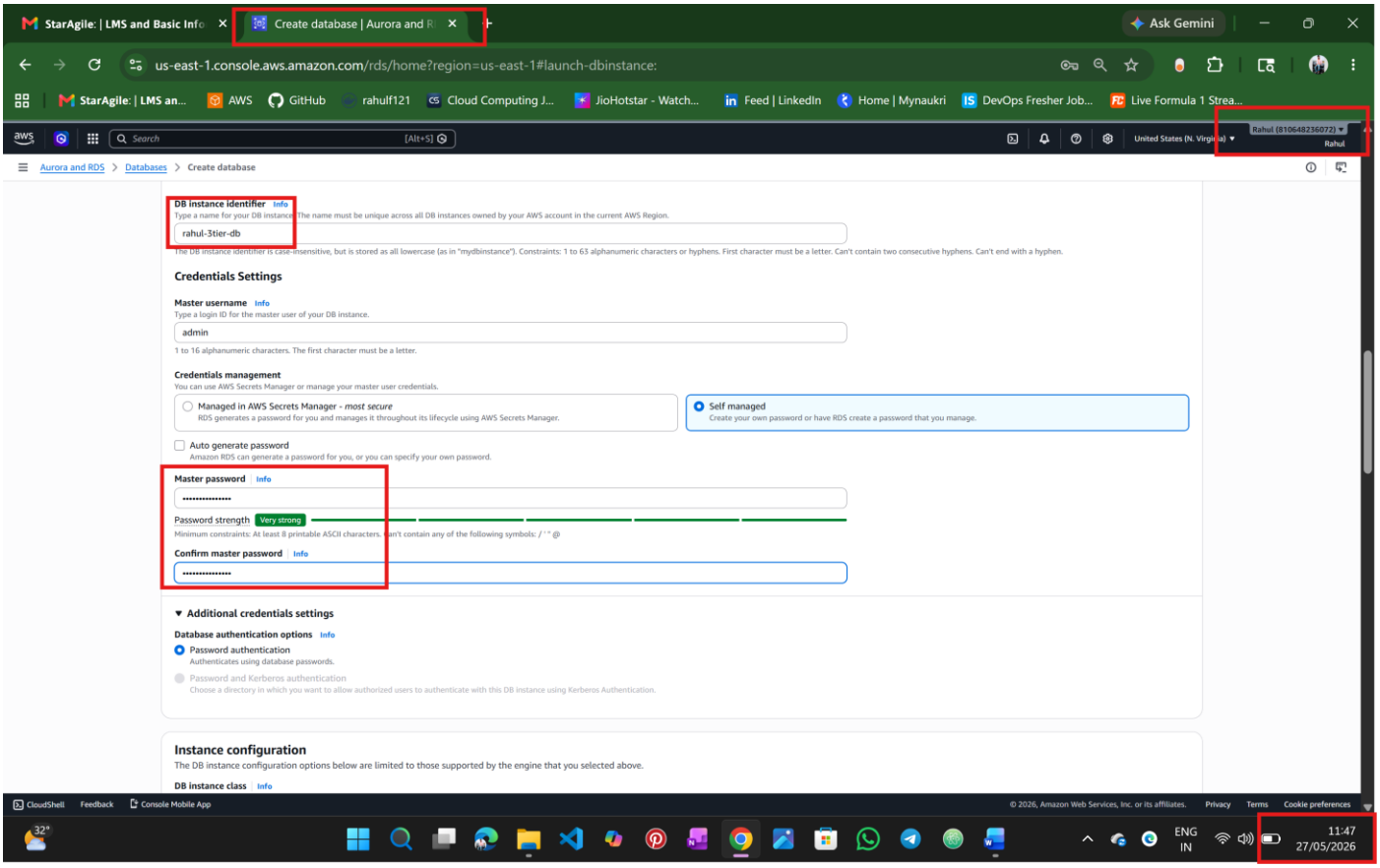

us-east-1.console.aws.amazon.com/s3/buckets/rahul-3tier-frontend?region=us-east-1&tab=properties
Amazon S3 > Buckets > rahul-3tier-frontend
Successfully edited static website hosting.
Transfer acceleration
Use an accelerated endpoint for faster data transfers. Learn more
Transfer acceleration
Disabled
Edit
Object Lock
Store objects using a write-once-read-many (WORM) model to help you prevent objects from being deleted or overwritten for a fixed amount of time or indefinitely. Object Lock works only in versioned buckets. Learn more
Object Lock
Disabled
Edit
Requester pays
When enabled, the requester pays for requests and data transfer costs, and anonymous access to this bucket is disabled. Learn more
Requester pays
Disabled
Edit
Static website hosting
Use this bucket to host a website or redirect requests. Learn more
Edit
We recommend using AWS Amplify Hosting for static website hosting
Deploy a fast, secure, and reliable website quickly with AWS Amplify Hosting. Learn more about Amplify Hosting or View your existing Amplify apps
Create Amplify app
S3 static website hosting
Enabled
Hosting type
Bucket hosting
Bucket website endpoint
When you configure your bucket as a static website, the website is available at the AWS Region-specific website endpoint of the bucket. Learn more
http://rahul-3tier-frontend.s3-website-us-east-1.amazonaws.com
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
15:41
26/05/2026



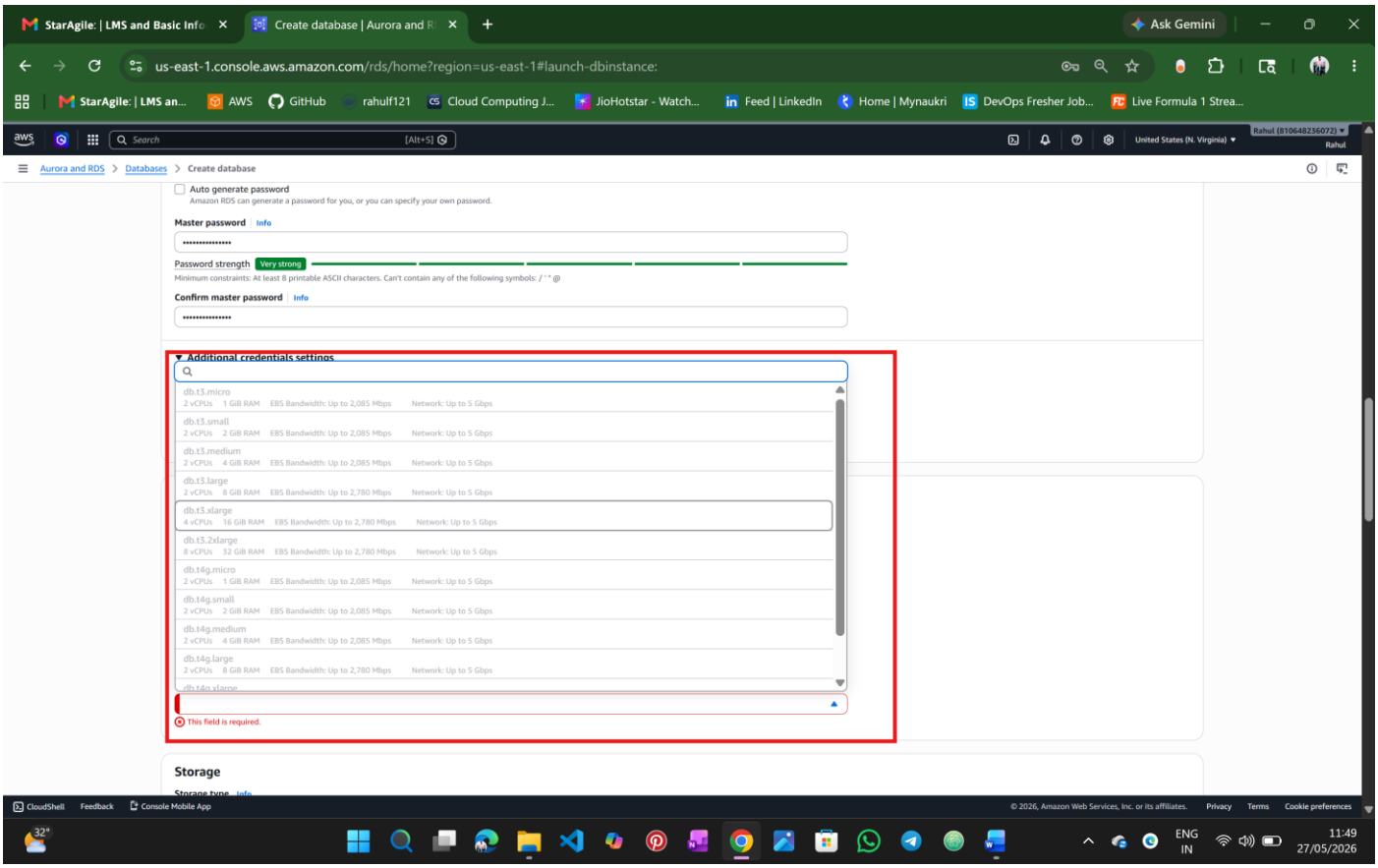
Step 4: Create RDS MySQL Database

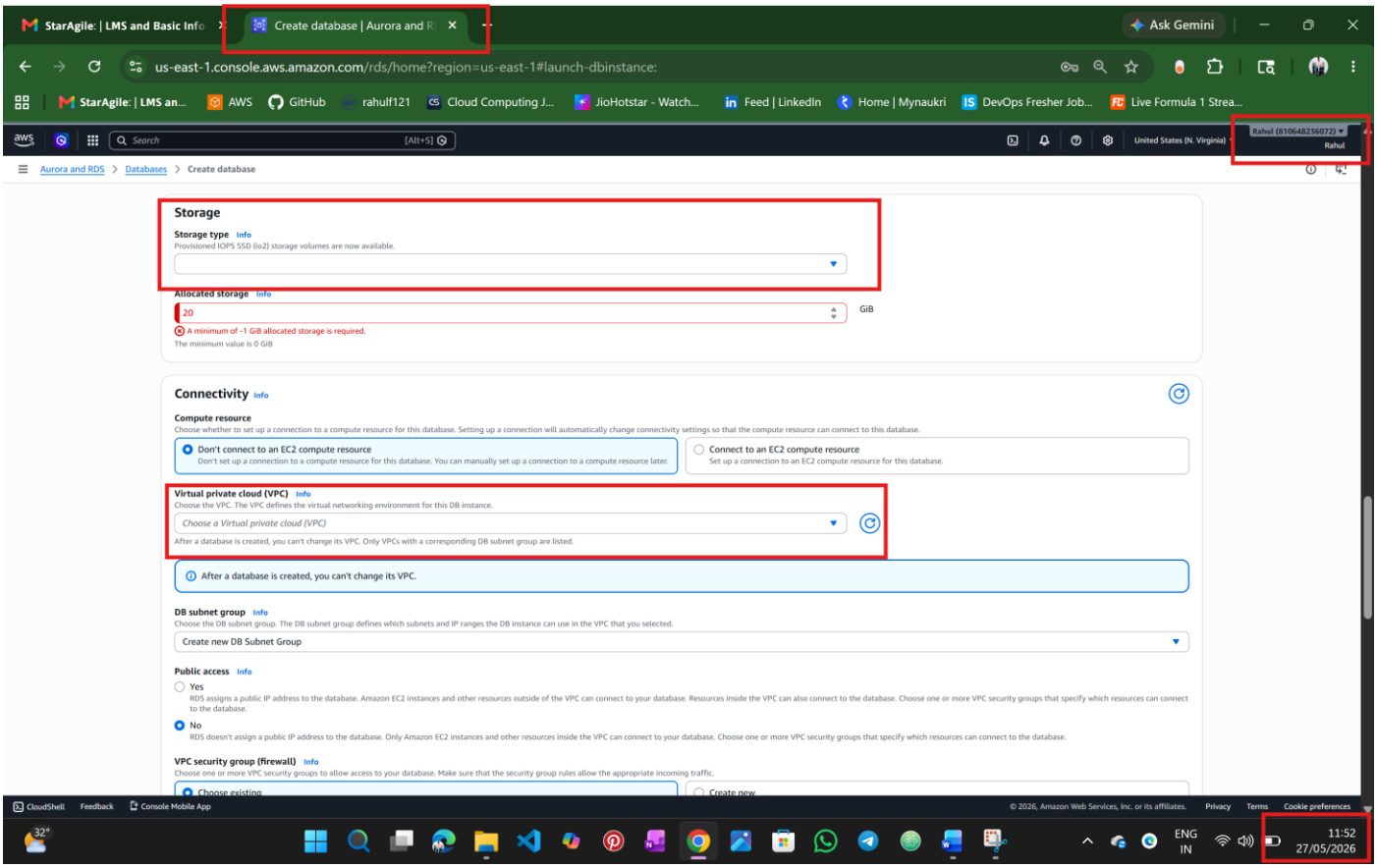
AWS Console → RDS → Create Database





Note: to select the Instance type and VPC (can't able to select the default also), so without implementation will move theoretically





Configuration

Setting	Value
Database Engine	MySQL
Template	Free Tier
Deployment	Single-AZ
DB Name	transactionsdb
Username	admin
Public Access	No
Security Group	database-sg

Database deployed in Private Subnet only

Step 5: Create Database Table

Connect from EC2:

mysql -h RDS_ENDPOINT -u admin -p (after RDS create it will show this end point)

it will ask for which DB

USE transactionsdb;

Create table:

```
CREATE TABLE transactions (
    id INT AUTO_INCREMENT PRIMARY KEY,
    amount INT,
    description VARCHAR(255)
);
```

Verify:

SHOW TABLES;

Step 6: EC2 Backend Server

Configure Database Connection(we use VIM command to edit the DB credentials)

Vim TransactionService.js

```
host: "RDS_ENDPOINT",
user: "admin",
password: "PASSWORD",
database: "transactionsdb"
```

Start Backend Server

node index.js

Expected:

AB3 backend app listening at http://localhost:4000

Test Backend

http:// 3.80.148.145:4000/health

Expected:

"This is the health check"

Step 7: Connect Frontend to Backend

src/components/DatabaseDemo/DatabaseDemo.js

Replace:

'/api/transaction'

With:

'http:// 3.80.148.145:4000/transaction'

Rebuild Frontend

npm run build

Reupload Frontend Build to S3

Upload updated build contents again.

Final Testing

Open S3 website endpoint.

Test:

- Add transaction
- View transactions
- Delete transactions
- Refresh page

Verify:

- Data stored in RDS
- Frontend communicates with backend
- Backend communicates with database

Conclusion

This project successfully demonstrates a traditional AWS 3-tier architecture deployment using native AWS services.

The application is fully functional with:

- Frontend hosted on S3
- Backend running on EC2
- Database hosted on RDS
- Secure networking using VPC and Security Groups
- End-to-end communication between all tiers